[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

May-26-0025

CSPC-613 (Compiler Design)

B.Tech-6th (NEP) CS&E, AI&MI (Session 2022-26)
B.Tech.-6th NEP CSE/CSE (AI&DS)CSE (AI&ML)

Time : 3 Hours Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

Section A

1. (a) Define compiler, interpreter, and highlight their differences with suitable examples. (6)

   (b) Explain the various phases of compilation with neat diagrams. (6)

2. (a) Describe the role of Lexical Analyzer and explain input buffering techniques. (6)

   (b) Explain the construction of DFA from Regular Expression with a suitable example. (6)

Section B

3. (a) What is ambiguity in grammars? Discuss methods to remove ambiguity. (6)

   (b) Explain Recursive Descent Parser with a working example. (6)

4. (a) Differentiate between LL(1) parser and LR(0) parser with examples. (6)

   (b) Explain the role of YACC tool in syntax analysis. (6)

Section C

5. (a) Explain syntax-directed definitions and evaluation orders. (6)

   (b) Construct a Three Address Code for the expression: A = (B + C) * (D – E). (6)

6. (a) Draw and explain a syntax tree for the statement: while (a < b) {a = a + 1;}. (6)

   (b) Write short notes on type checking and translation of declarations. (6)

Section D

7. (a) Explain the need for runtime memory management in detail. (6)

   (b) Discuss stack allocation and heap management with diagrams. (6)

8. (a) What are the principal sources of code optimization? Explain with examples. (6)

   (b) Explain DAG representation and its role in optimization. (6)

Section E (Compulsory)

9. (a) Define bootstrapping in compiler design.

   (b) Differentiate between Top-down and Bottom-up parsing.

   (c) What is peep-hole optimization? Give an example.

   (d) Mention any two issues in code generation.

   (e) Write the advantages of minimizing DFA.

   (f) Explain various types of peep-hole optimizations. (6×2=